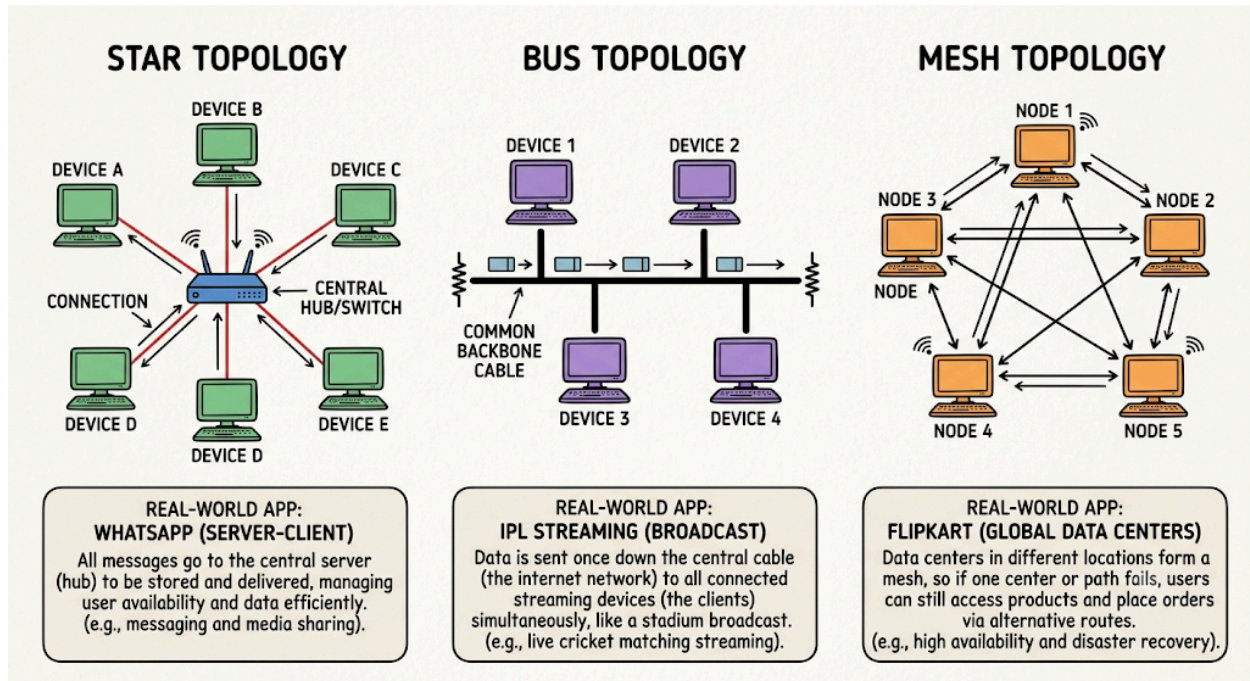


NETWORK TOPOLOGIES

Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

ANS:



Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

ANS:

Topology	Pros	Cons	Best Use Case	App Example
Star	Easy setup, isolated failures, scalable, simple troubleshooting	Central hub single point of failure, high cabling cost	Client-server apps	WhatsApp / Zomato

Bus	Cheap, minimal cabling, easy small setup, no central node	Cable failure drops network, high collisions, bad under heavy load	One-way broadcasts	IPL Streaming
Ring	No collisions, equal access, handles load well, no central node	Single node breaks ring, adding nodes disrupts network, shared bandwidth	Token networks / SONET fiber	Paytm Pipelines
Mesh	Fully redundant, zero downtime, high security, handles heavy traffic	Extremely expensive, complex setup, high maintenance	Critical infrastructure	Instagram / AWS

Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

ANS:

For a multiplayer mobile game like PUBG or Free Fire, a hybrid **Star-Mesh Topology** (Client-Server Star locally, Mesh globally) is the optimal choice:

- **Speed (Latency):** Players connect via a **Star topology** directly to regional game servers, minimizing ping and keeping real-time player movements synchronized without peer-to-peer lag.
-
- **Reliability:** Regional servers are interconnected in a **Mesh topology**, allowing live match data, player stats, and inventory databases to sync instantly with zero single points of failure.
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- **Cost:** Higher initial infrastructure and server hosting costs, but essential for supporting millions of concurrent low-latency connections globally.

Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.

ANS:

Recommendation: Star Topology

Connect all 8 warehouse PCs to a central 16-port Ethernet Switch.

Diagram

```

[PC 1] [PC 2] [PC 3] [PC 4]
  |     |     |     |
[===== Central Switch =====]
  |     |     |     |
[PC 5] [PC 6] [PC 7] [PC 8]

```

Reasoning

- **Low Cost & Easy Expansion:** Cheap to set up; adding new inventory stations just requires plugging extra cables into the switch.
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- **Isolated Failures:** If one PC or cable breaks, the other 7 order-processing stations keep running normally.